

Problem Set 1

Theme: Conservation genetics of an endangered felid
Topics tested: HWE, F_{IS} , LD, recombination, F_{ST} , effective population size
Due: End of Week 4
Total points: 100
Estimated time: 4–5 hours

Biological Scenario

The **Anatolian leopard** (*Panthera pardus tulliana*) is a critically endangered subspecies with fewer than 15 confirmed individuals remaining in Turkey. A captive breeding programme has been running for 12 years at two separate facilities—**Facility North (FN)** and **Facility South (FS)**—each founded from wild-caught animals with no subsequent exchange of individuals.

The programme manager has two concerns she needs you to address with genetic data:

1. She suspects that animals within each facility may no longer be mating randomly, because handlers have observed that dominant males monopolise most matings.
2. She wants to know whether 12 years of separate management has made the two facilities genetically divergent, and whether merging them for a joint breeding programme would be safe.

You have genotyped all available individuals at two SNP loci from a panel developed for *Panthera* species.

Data file: `leopard_genotypes.tsv`

The file contains 80 individuals (40 per facility) genotyped at two SNP loci:

- **Locus 1** — A/T SNP, column Locus1_AT
- **Locus 2** — C/G SNP, column Locus2_CG

Each row is one individual. The two loci are **200 kb apart** on the same chromosome, in a region with a recombination rate of **0.5 cM/Mb**. Assume a leopard generation time of **3 years** (approximately 4 generations since the facilities were founded).

Problem 1 Diagnosing the Breeding System [35 pts]

The programme manager's first concern is that dominant male monopolisation is reducing the effective number of breeders. Before advising her, you need to characterise mating patterns in each facility independently.

- (a) [12 pts] Analyse the Locus 1 genotype data from **both facilities** to determine whether mating within each is consistent with random mating. Perform the appropriate statistical test, report your results clearly, and state what you conclude about each facility. Do your results support the programme manager's concern? Are the two facilities showing the same pattern?

Reminder — Chi-square test for HWE:

$$\chi^2 = \sum_i \frac{(E_i - O_i)^2}{E_i}, \quad E(AA) = \hat{p}^2 N, \quad E(AT) = 2\hat{p}\hat{q} N, \quad E(TT) = \hat{q}^2 N$$

Critical value: $\chi^2_{1\text{df}} = 3.84$ at $p = 0.05$. The inbreeding coefficient $F_{IS} = (H_S - H_I)/H_S$ quantifies the direction and magnitude of departure from random mating.

- (b) [10 pts] The pattern you observe in one or both facilities could have more than one biological cause. The programme manager assumes it is caused by male monopolisation of matings. Evaluate this explanation: is it consistent with your results? Propose at least **one alternative** biological explanation that would produce the same genetic signature through a completely different mechanism. How would you distinguish between the two explanations if you had access to studbook records (records of which animals produced which offspring)?
- (c) [13 pts] You find a striking difference in F_{IS} between the two facilities, despite the fact that they were founded from the same wild source population. The programme manager asks: *"Does this mean the animals in one facility are more inbred?"*
Write a careful response to her question (5–6 sentences) that correctly explains what F_{IS} measures, what it does and does **not** tell you about inbreeding in the classical pedigree sense, and what additional data you would need to make a definitive statement about inbreeding in these animals.

Problem 2 Should the Facilities Be Merged? [35 pts]

The programme manager's second concern is whether 12 years of separate management has made the two facilities genetically distinct. A colleague has warned her that "if F_{ST} is high, merging the populations could cause outbreeding depression."

- (a) [10 pts] Calculate the relevant statistic(s) to quantify genetic differentiation between the two facilities at Locus 1. Interpret your result using established benchmarks and state clearly what you would advise the programme manager based on this analysis.

Reminder — F_{ST} :

$$H_T = 2 \bar{p} \bar{q}, \quad \bar{H}_S = \frac{H_{S, FN} + H_{S, FS}}{2}, \quad F_{ST} = \frac{H_T - \bar{H}_S}{H_T}$$

Benchmarks: <0.05 little, 0.05–0.15 moderate, 0.15–0.25 great, >0.25 very great differentiation.

- (b) [12 pts] A reviewer of your report argues: "A single locus is not sufficient to draw conclusions about population differentiation. Your F_{ST} estimate is unreliable." Respond to this criticism in detail. Explain why the reviewer is correct from a statistical standpoint, describe how the reliability of an F_{ST} estimate depends on the number of loci and the number of individuals sampled, and state what minimum dataset you would recommend before making a management decision. No additional calculations are needed—this is a reasoning question.
- (c) [13 pts] Examine the two-locus data from Facility North. Is there evidence of non-random association between Locus 1 and Locus 2? Quantify this using an appropriate statistic and interpret the result. Then address the following: given that the two loci are 200 kb apart in a region with a recombination rate of 0.5 cM/Mb, and given that FN has been a closed population for approximately 4 generations, is the level of association you observe **expected, surprising, or ambiguous**? Show your reasoning using the recombination decay formula and interpret the result in the context of the facility's short population history.

Reminder — LD statistics and decay:

$$D = f(A-C) - f_1(A) f_2(C), \quad r^2 = \frac{D^2}{f_1(A) f_1(T) f_2(C) f_2(G)}, \quad D_t = D_0 (1 - r)^t$$

Double heterozygotes (AT/CG) contribute 0.5 to each of the two possible haplotype pairings when estimating haplotype frequencies.

Problem 3 Communicating Uncertainty [30 pts]

The programme manager must present findings to a government committee that will decide whether to fund the merging of the two facilities.

- (a) [15 pts] She shows you a draft statement she intends to present to the committee:

*“Our genetic analysis **proves** that the two leopard populations are still similar and can be **safely** merged.”*

Identify every claim in this statement that is scientifically unjustified based on your analyses, and rewrite it as a scientifically accurate statement that still makes a clear recommendation. Your rewrite should be accessible to a non-specialist audience but must not overstate what the data show.

- (b) [15 pts] The committee asks: *“If the animals in Facility North are not mating randomly, does that mean their genetic diversity is decreasing faster than it should be?”*

Answer this question rigorously in the context of effective population size and genetic drift. Your answer should explain the concept of N_e to a non-specialist, connect non-random mating and F_{IS} to the rate of genetic drift, and make a concrete statement about the long-term genetic consequences for the captive population if the current breeding pattern continues. (6–8 sentences)